Week-2

1.Swap Two numbers (call by value &call by reference

```
#include <iostream>

using namespace std;


void swapValue(int a, int b)

{

   int temp = a;

   a = b;

   b = temp;

}


void swapReference(int &a, int &b)

{

   int temp = a;

   a = b;

   b = temp;

}


int main()

{
```

```
    int x = 10, y = 20;

    swapValue(x, y);
    cout << "After call by value: " << x << " " << y << endl;

    swapReference(x, y);
    cout << "After call by reference: " << x << " " << y << endl;

    return 0;
}
```

2.Add 1D Array using elements.

```
#include <iostream>

using namespace std;

int main()

{

    int n, sum = 0;

    cout << "Enter number of elements: ";

    cin >> n;
```

```
    int arr[n];

    cout << "Enter elements: ";

    for(int i = 0; i < n; i++)

    {

        cin >> arr[i];

        sum = sum + arr[i];

    }

    cout << "Sum = " << sum;

    return 0;

}
```

3.Read 1D Array Elements Using pointer

```
#include <iostream>

using namespace std;
```

```
int main()
{
    int arr[5];
    int *ptr = arr;
    cout << "Enter 5 elements:\n";
    for(int i = 0; i < 5; i++)
    {
        cin >> *(ptr + i);
    }
    cout << "Array elements:\n";
    for(int i = 0; i < 5; i++)
    {
        cout << *(ptr + i) << " ";
    }
```

```
  return 0;

}
```

4.Read 1D Arayy Using DMA

```
#include <iostream>

using namespace std;

int main()

{

   int n;

   cout << "Enter number of elements: ";

   cin >> n;

   int *arr = new int[n];

   cout << "Enter elements:\n";

   for(int i = 0; i < n; i++)
```

```
    {
        cin >> arr[i];
    }
    cout << "Array elements:\n";
    for(int i = 0; i < n; i++)
    {
        cout << arr[i] << " ";
    }
    delete[] arr;
    return 0;
}
```

5. Function over loading

```
#include <iostream>
using namespace std;
```

```
int add(int a, int b)

{

    return a + b;

}

int add(int a, int b, int c)

{

    return a + b + c;

}

int main()

{

    cout << "Sum of two numbers = " << add(5, 10) << endl;

    cout << "Sum of three numbers = " << add(5, 10, 15);

    return 0;

}
```